

## 1. PROBLEM STATEMENT

Weightlifting is a widely practiced form of exercise, but it presents challenges for both beginners and experienced participants. Many individuals struggle with maintaining proper form, understanding gym equipment, and selecting appropriate weights or workout plans. These challenges often result in injuries, frustration, and decreased motivation for those seeking safe and effective training.

### 1.1 Need Statement

Improper form during weightlifting contributes to injuries, with approximately 970,000 weightlifting-related injuries reported in the United States between 1990 and 2007 [1]. Many beginners find gyms intimidating because of the complexity of the equipment and the absence of clear guidance [2]. Selecting appropriate weights and exercises often requires significant research, which can overwhelm new users [3]. Current fitness tools, including mobile applications and basic wearables, provide data such as steps, calories, or repetitions, but they lack the ability to deliver real-time feedback on form [4]. This gap leaves individuals at risk of injury and limits their ability to train effectively without professional supervision.

### 1.2 Objective Statement

The objective of this project is to design a portable system (all detachable sensors located in a tote-around case) that improves weightlifting safety and effectiveness. This comprehensive system - dubbed the *Repetition. Efficiency. Precision (R.E.P.) Tracker* - utilizes advanced accelerometer tracking, and easy-to-use heart rate, oxygen, and temperature sensors to sense the user's current performative state (e.g. weightlifting form and workout intensity); the system's user-feedback, user-interface, and a camera recorder provide quality assessments through vocal (text-to-speech programming) and visual (recordable camera feed) outputs to better recommend exercise and body adjustments to the user. These guided outputs and metrics are based on the data processed from the accelerometer and the aforementioned sensors. To summarize, the R.E.P Tracker system analyzes body motion, counts repetitions and sets, and communicates feedback in real time to guide proper technique and monitor long-term progress so that the user can have a stronger, safer, and healthier fitness journey.

### 1.3 Background and Related Work

Motion tracking and camera-based monitoring are widely used in fitness technology. Consumer wearables such as Fitbit track steps and calories but do not analyze movement patterns in detail [4]. High-precision systems such as Vicon deliver accurate motion capture for research and clinical applications, but they are expensive and impractical for general gym use [5]. Smart home gym products such as Mirror and Tempo Studio provide form feedback through large interactive screens, but their high cost and subscription requirements reduce accessibility [6]. Existing patents on motion-tracking technologies demonstrate general approaches to athletic monitoring but do not directly address affordable, portable weightlifting feedback systems [7]. This project focuses on bridging that gap by combining portability, low cost, and real-time form analysis in a compact solution.

## 2. Design Requirement Specifications

### 2.1 Marketing Requirements

1. The R.E.P. Tracker should provide long-lasting battery life so users can complete multiple workout sessions or tests without needing frequent recharging.
2. The user interface should be easy to read from a reasonable workout distance and remain visible in typical gym lighting conditions.
3. The system should accurately track movement and securely attach to standard gym equipment without slipping or losing data.
4. The device should provide dependable, real-time feedback on user performance and vital statistics to help users track progress safely and effectively.
5. Users are provided with the ability to adjust audio cues to match their environment and preferences while maintaining safe listening levels.
6. The system should capture clear video footage that allows users and trainers to review exercise form and techniques effectively.
7. Smooth and responsive wireless communication will be provided to the user to ensure that performance data and accurate feedback are delivered in real time.
8. The R.E.P. Tracker will function in typical workout environments, including variable temperatures and humidity levels found in gyms or outdoor settings.

Figure 2-1 shows an objective tree for the R.E.P. Tracker.

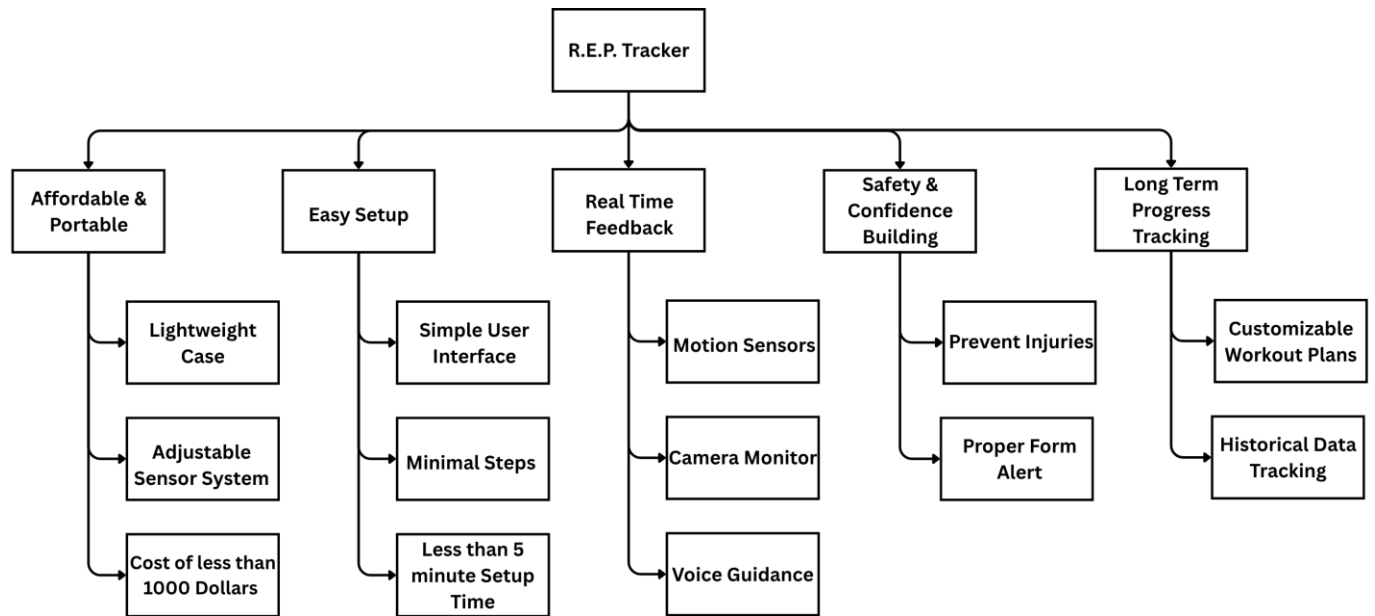


Figure 2-1: R.E.P. Tracker Objective Tree

## 2.2 Engineering Requirements

As seen below, Table 2-2 outlines the Engineering Requirements identified for the REP Tracker in congruence with the Marketing Requirements.

Table 2-2: Engineering Requirements

| Marketing Requirements           | Requirements                                                                                                                               | Reasoning                                                                                                                                                       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.Power                          | Main battery must charge 3 2000 mAh batteries for 2 cycles of 2000 mAh.<br><br>2000 mAh batteries must last 4 hrs minimum under full load. | A sustained battery life is needed to use the system for an extended period for exercising and testing purposes.                                                |
| 2. Display                       | Data trending readable from a distance of 4ft<br><br>Adjustable brightness capable of at least 400nits                                     | The display needs to be readable in average lighting conditions in gym or exercise environments.                                                                |
| 3. Motion Tracking               | Accelerometer/Gyroscope/Magnetometer data accuracy 2% error maximum<br><br>MT must latch on common barbell type and materials              | Accuracy and stability minimize glitching and maximize compatibility with relevant/popular exercise equipment (e.g. barbells and similarly configured objects). |
| 4. Performance Monitoring        | Accuracy: error <2% oxygen measurement, error <10% temperature readings, error <10% heart rate data<br><br>Must be wearable by the user    | The system needs accurate vitals for trending purposes and a solid fitting sensor for reliability.                                                              |
| 5. User Feedback                 | Adjustable audio output not exceeding 85dBA                                                                                                | Audio requires safety parameters and adjustable volume for auditorial environmental adaptation.                                                                 |
| 6. Camera Feed                   | Minimum of 1080p at 30fps recording quality                                                                                                | The video recorded needs a minimum quality for analysis and legibility.                                                                                         |
| 7. Wireless Communication        | BLE latency under 300ms<br><br>Range of up to 10m                                                                                          | The device needs to maintain a consistent, real-time data stream.                                                                                               |
| 8. Standard Operating Conditions | 0°C to 40°C<br><br><85% humidity                                                                                                           | The common conditions in which the system will need to operate.                                                                                                 |

Now, to elaborate on the specific engineering requirements for the REP Tracker system. Starting with power – the main battery (15,000mAh), fully charged, should be able to charge full load (three 2000mAh batteries) for at least two charging cycles. Each 2000mAh battery should be able to last at least four hours fully charged.

Continuing to Motional Tracking (MT), the accuracy of data processed and transmitted by the BNO055 module must be within 5% of error. Additionally, the MT housing must be able to be latched onto any barbell of any material given the diameter of the “sleeve” (the part of the barbell which the plate

slides onto) is 2 inches in diameter [8]. Both the accuracy of MT metrics and the conformability of the module are essential to ensure consistent and reliable data.

The Performance Monitoring (PM) subsystem's accuracy should be less than or equal to 2% error for oxygen monitoring [9], 10% for temperature readings, and 10% for heart rate monitoring. These metrics should be within acceptable accuracy range so that tracking and trending is precise and reflective of actual conditions measured. To be specific, there is legality governing oxygen readings [9].

For the Camera Feed subsystem, the captured video feed must have, at minimum, a quality of 1080p at 30fps for form analysis. These video quality metrics maintain a readable recording so that the user can analyze the details presented through the Camera Feed.

Furthermore, the User Feedback subsystem must not exceed 85dBA in volume [14] and must have an adjustable volume output with a visual display through the User Interface. This allows the user to adjust the output volume to the appropriate level given surrounding auditorial conditions. This also maintains NIOSHA hearing safety standards for the user.

Subsequently, the User Interface must have an adjustable screen brightness capable of at least 400nits [15]. The data trending displayed (reps, sets, timing, suggestions, etc...) needs to be legible from up to 4ft. This allows the user to easily view relevant metrics given the average distance from the display and regardless of ambient lighting conditions.

Finally, the requirements for communication protocols - the wireless Bluetooth communication must maintain a latency of under 300ms and have a typical operating range of up to 10m. This requirement ensures smoother displayed metrics and maintains glitch resistant communication between different parts of the REP Tracker system.

Comprehensively, the whole system should optimally operate within 0°C to 40°C and up to 85% humidity. This allows for operation in most gym or exercise settings.

After the previously discussed Engineering Requirements are the constraints of the REP Tracker.

## 2.3 Constraints

Table 2-3 has constraints related to the R.E.P. Trackers development process that goes over cost, cime, weight, required space and lighting, range, and safety. These constraints are listed below:

Table 2-3: Constraints

| Type     | Name | Description                                                     |
|----------|------|-----------------------------------------------------------------|
| Economic | Cost | The project budget is limited to \$700, including both hardware |

|                   |                    |                                                                                                            |
|-------------------|--------------------|------------------------------------------------------------------------------------------------------------|
|                   |                    | development and backup components.                                                                         |
| Economic          | Time               | The project must be completed by the end of April.                                                         |
| Manufacturability | Weight             | The full system (all components) needs to weigh under 10kg (20.05 lbs).                                    |
| Functionality     | Space and lighting | The design must work in gyms with limited space and variable lighting (300 – 400 Lux) [9]                  |
| Functionality     | Range              | The clear communication range for Bluetooth is up to 30ft.                                                 |
| Health and Safety | Safety             | Must ensure shock protection for magnetic power connectors and overall system (and/or sufficient warning). |

The \$700 budget in the cost constraint refers to the limited funds for parts that we can purchase. This ties into the time constraint as well; the budget limits cheaper parts in certain aspects and the deadline forces prioritization of tasks.

Manufacturability refers to the weight of the R.E.P. Tracker. Making sure the system's weight does not exceed 10kg is important as too much weight becomes a hindrance. Additionally, the system being cumbersome could cause further issues beyond user convenience as significant heaviness could disrupt components during dynamic or sustained movement.

Functionality is one of the most impactful constraints as this outlines the necessary space and lighting for using the device. Room to operate is an essential consideration as the system does not need a large space but only needs enough leeway to function without disrupting the flow of the gym environment. Stable Bluetooth connections are solid up to a distance and that distance factors into the set-up process for the device (the fundamental nature of the transmitting waves limits the R.E.P. Tracker wireless range).

In Health and Safety it goes over shock protection and adequate warnings, this is extremely important as the device needs to have precaution in making sure no accidental shock or harm can come to the user, Functionality ties in again with lighting - which will ensure the device can be safely prepared ; thereby, preventing the user from accidentally exposing themselves to danger.

Constraints are often accompanied by related standards or the standards themselves create constraints (legality, policy, etc...). Next are the standards relating to the R.E.P. Tracker.

## 2.4 Standards

Table 2-4 References standards that are applicable to R.E.P. Tracker.

Table 2-4: Standards

| Standard    | Relevance                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------|
| FCC Part 15 | Ensures the device does not cause harmful interference with other wireless devices in gyms [11]. |

|                               |                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------|
| IEEE 802.15.1/802.15.4        | Define Bluetooth communication protocols, ensuring interoperability and low-latency data transfer [12]. |
| IEC 62368-1/OSHA/NFPA 70(NEC) | Provides safety standards for electronic devices to reduce risks of electrical hazards [13].            |
| FDA-2007-D-0205               | Legality requires $\pm 3\%$ minimum error percent for measuring blood oxygen for a device [9].          |

Following the guidelines of these standards our system must not interfere with local devices in proximity to its physical location or the network it is connected to. We will also avoid any dangerous or malicious software from being sent or received by the REP Tracker device [11].

Furthermore, our device will be prompt in communicating with the modules separately via Bluetooth. Communication has a typical range of up to 10 meters [12]. Another vital and important standard that we will keep is to provide safe use for all components containing electric components. All components that contain any charge must be properly insulated and housed. This is to ensure the safety of all users during the use of our device [13].

Measuring blood oxygen levels is vital to a person's health. The standards for devices that measure blood oxygen are very strict and cannot have more than a three percent error range in its calculations for devices located on the arm. These medical standards are held in all fields of products distributed for health and wellness. We plan to keep these same medical standards to ensure that the development of our device does not become stagnant [9].

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